

Embedded Systems

A practical bridge between electronics and software: boards, processors, peripherals, sensors and real-world systems.

Recommended learning order

1. Microcontroller basics
2. GPIO and peripherals
3. Sensors and actuators
4. Arduino / Raspberry Pi prototyping
5. Build and debug small embedded projects

Direct YouTube resources

Source	What to use it for	Open
Ben Eater	Build a computer from scratch — strong low-level hardware intuition.	YouTube →
Phil's Lab	Embedded systems / hardware design tutorials.	YouTube →
Paul McWhorter	Arduino tutorial series — beginner-friendly embedded prototyping.	YouTube →
Core Electronics	Raspberry Pi Workshop — practical embedded-computing introduction.	YouTube →

Tip: open the YouTube → buttons directly from the PDF. This roadmap is based on the supplied Hardware Learning Roadmap, with Computer Networks added as a separate course because you specifically requested it.

Prepared as a separate topic-wise learning PDF.